

Separation of azeotropic mixtures using high ionicity ionic liquids based on 1-ethyl-3-methylimidazolium thiocyanate

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ABSTRACT

The separation of azeotropic mixtures, so that their components can be reused in productions cycles, is of particular interest in modern sustainable chemical plants. In the last decade, ionic liquids have shown to possess the ability to act as extraction solvents and have been emerging as a greener alternative to organic solvents in the separation of azeotropes. Recently, high ionicity ionic liquids (HIILs) have been developed through the addition of inorganic salts to an ionic liquid, as well as by mixing two or more ionic liquids. In this work, a high ionicity ionic liquid, prepared by mixing 0.1 of 1-ethyl-3-methylimidazolium thiocyanate ($[C_2MIM][SCN]$) and 0.9 of $[C_2MIM][C_2SO_4]$ was used to separate the azeotrope ethanol + heptane. Liquid–liquid equilibria for the two ternary systems: ethanol + heptane + $[C_2MIM][SCN]$ and ethanol + heptane + HIIL was measured at 298.15 K and 0.1 MPa. Both the selectivity and the distribution coefficient were used in the assessment of the extraction solvent feasibility and a correlation of these parameters with the ionicity of the solvent was established.

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1. Introduction

Over the last decade, the scientific community has been paying more attention to novel processes based on greener technologies, due to the increasing concern about environmental issues, as well as the establishment of new regulations. In many areas of industry, solvent mixtures accumulate due to recycling difficulties. Their separation into pure components is necessary so that they can be reused. However, many of these solvent mixtures contain azeotropes, which are special mixtures whose separation is still an issue in chemical engineering.

Several potential processes, such as azeotropic distillation, extractive distillation and liquid–liquid extraction, have been used to selectively remove one compound of an azeotropic mixture [1]. The distillation techniques involve high temperatures or high pressures to reach a single fluid phase system, leading to high energy consumptions. Alternatively, liquid–liquid extraction takes place at room temperature and atmospheric pressure and therefore, economic issues related to the energy consumed due to vaporisation do not exist. The conventional extraction solvents used in industry are organic solvents, such as sulfolane [2] and tetraethylene glycol, which, due to their generally volatile, toxic and/or flammable nature, require further investments and energy for their recovery. To this

end, ionic liquids (ILs) have emerged as an appealing option, since their use and recovery involve fewer steps, rendering less energy intensive processes [3]. ILs have already demonstrated their high separation efficiency in breaking azeotropes including outstanding examples such as ethanol + water, alcohol + alkane mixtures [1] and aliphatic + aromatic mixtures [4,5].

Over the past few years, the concept of ionicity of ILs has been debated by several authors [6]. It has been shown that ionicity is closely related to the occurrence of ion pairs, the formation of aggregates and the existence of ionic networks. Thus, an increase in ionicity can be translated into ion's "availability" to participate in solvation schemes with other molecular components present in the system. The simplest way to "measure" ionicity is based on the Walden rule [7–9], where the relationship between the fluidity and the ionic conductivity is evaluated through Eq. (1) as follows:

$$\Lambda_m \times \eta = c \quad (1)$$

where Λ_m is the molar conductivity, η is the viscosity and c is a temperature dependant constant. On a log–log plot of the molar conductivity versus the fluidity, usually known as the Walden Plot, this rule predicts a straight line that passes through the origin for a fully dissociated fluid [9], the ideal Walden line, usually represented using data of an aqueous solution of KCl. Angell and co-workers [7–9] showed that most ILs fall below this line, acknowledging the strong interactions between anions and cations. However, previous work from our laboratory established that ionic liquid's ionicity could be improved by the dissolution of

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simple inorganic salts (IS) in their milieu [10,11]. In the same line of thought, the use of a mixture of ILs to increase the solvent ionicity might allow the “design” of a new solvent, with specifically fine-tuned properties including ionicity. IL binary mixtures, such as *N*-butylpyridinium bis(trifluoromethylsulfonyl) imide ([bpy][NTf₂]) and *N*-butylpyridinium tetrafluoroborate ([bpy][BF₄]) [4], have already been tested as extraction solvents to separate toluene from *n*-heptane. The IL mixtures used in this paper presented an ideal behaviour, unlike one of the systems studied in this work, which led to the question of the effect of using a non-ideal solvent to separate azeotropes. A possible correlation between the ionicity of fluid and its extraction efficiency for azeotrope separation is here researched for the first time.

The main goal of this work is to verify a possible relationship between the ionicity of a mixture of ionic liquids and its effectiveness in the liquid–liquid extraction for breaking an azeotrope. For that purpose, and based on our previous studies, the thermophysical properties of three systems based on 1-ethyl-3-methylimidazolium thiocyanate, [C₂MIM][SCN], combined with an inorganic salt, ammonium thiocyanate, [NH₄][SCN], and two ionic liquids, 1-ethyl-3-methylimidazolium ethyl sulfate, [C₂MIM][C₂SO₄], and 1-ethyl-3-methylimidazolium dicyanamide, [C₂MIM][DCA], were measured in a wide range of temperatures and at 0.1 MPa and their ionicity was evaluated. The second part of the work consists on using both the pure [C₂MIM][SCN] and the system with the largest ionicity from the three mixtures studied before, as extraction solvent for the ethanol+heptane azeotrope. This azeotrope is formed in the production of oxygenated additives in petrol, typically present in processes to achieve the reduction of lead in gasoline and the separation of these azeotropic systems is necessary so that the separated parts can be re-used in production

cycles [12,13]. For this purpose, the binodal curves and tie-lines for each ternary system were obtained at 298.15 K and 0.1 MPa and the selectivities and distribution coefficients were determined in order to establish which solvent is more appealing for the extraction of ethanol by liquid–liquid extraction and thus conclude about the possible relationship between ionicity and efficiency.

2. Experimental

2.1. Materials

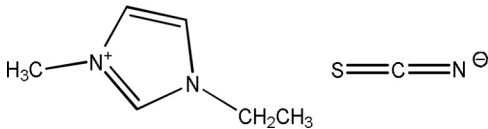
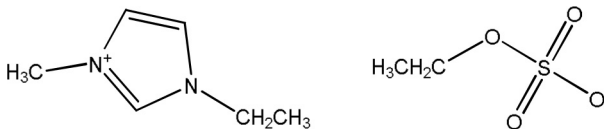
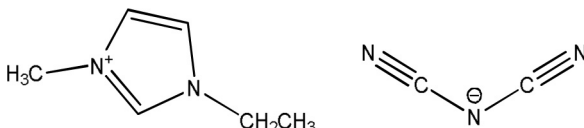
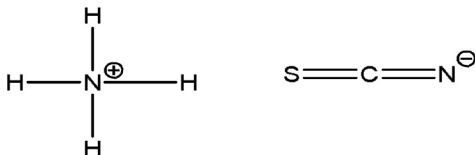
The chemical structures of the ILs and IS used in this study are illustrated in Table 1, including their source and purity. In order to reduce the water and other volatile substances contents, a moderate temperature (323.15 K) and vacuum (10^{−1} Pa) was applied to all samples for at least 48 h prior to their use. The final mass fraction of water was measured by Karl Fischer coulometric titration. All dried pure ILs and IL-based mixtures contained 900–1100 ppm of water. The components of the azeotropic mixture, heptane and ethanol, were purchased from Carlo Erba (99 wt%) and Panreac (99.5 wt%), respectively, and were used without any further purification.

2.2. Thermophysical properties

Density and viscosity. The measurements of density and viscosity were performed in the temperature range between 298.15 and 353.15 K at 0.1 MPa, using an automated SVM 3000 Anton Paar rotational Stabinger viscometer-densimeter. The temperature uncertainty was ±0.02 K. Further details on the equipment can be found elsewhere [13]. The highest uncertainty registered for the dynamic

Table 1

Name, acronym, chemical structure and supplier of ILs and IS used.

Compound name, acronym ILs	Chemical Structure	Supplier and purity (%)
1-Ethyl-3-methylimidazolium thiocyanate, [C ₂ MIM][SCN]		Fluka, 95%
1-Ethyl-3-methylimidazolium ethyl sulfate, [C ₂ MIM][C ₂ SO ₄]		Iolitec, 98%
1-Ethyl-3-methylimidazolium dicyanamide, [C ₂ MIM][DCA]		Iolitec, 98%
Inorganic salt		
Ammonium thiocyanate, [NH ₄][SCN]		Sigma-Aldrich, 97.5%

viscosity and density measurements was $\pm 2.19\%$ and $\pm 0.03\%$, respectively. The results presented are the average value of three replicas measured for the IL and each mixture.

Ionic conductivity. The measurements of ionic conductivities were performed using a CDM210 Radiometer Analytical conductivimeter in the temperature range between 298.15 and 323.15 K. The apparatus used for the measurements can be found elsewhere [10]. For the calibration of the equipment for each temperature, a certified 0.01 D KCl standard solution supplied by Radiometer Analytical was used. Each conductivity value was determined at least three times to ensure its reproducibility within 1% in absolute value.

Refractive index. The measurements of refractive indexes were performed using an automated Anton Paar Refractometer Abbemat 500 at the temperature of 298.15 K and pressure of 0.1 MPa. The absolute uncertainty of the equipment is ± 0.00005 . The refractive index of each sample was determined in triplicate with an absolute uncertainty of ± 0.00005 .

2.3. Ternary liquid–liquid equilibrium (LLE) measurements

Two different experimental procedures were used to study the ternary LLE of heptane + ethanol + solvent: the first was based on the direct determination of the binodal curve, by visually pinpointing the disappearance of the turbidity (cloud point); the second was centred on the determination of the tie-lines, by using the fittings densities and refractive indices with the composition along the binodal curve.

The determination of binodal curve was performed at 298.15 K and 0.1 MPa in a glass cell containing a magnetic stirrer. The cell was provided with an external jacket in which water circulated at the working temperature (± 0.01 K) using a Lauda E200 water thermostat and a Pt 100 probe coupled to a Keithley 199 System DMM/Scanner. The binodal curve of the heptane + ethanol + solvent ternary system was determined by preparing mixtures of the solvent (ionic liquid or HIL) and the inert (heptane) in the immiscible region. The mixture was placed inside of the jacketed cell and known quantities of ethanol were then carefully added, under constant stirring, until the ternary mixture became totally miscible, that is, the cloud point of the mixture was obtained. Afterwards, the density and the refractive index of those ternary mixtures were determined in triplicate.

For the determination of the tie-lines, about 10 mL of ternary mixtures of known composition within the immiscibility region were prepared in vials and were stirred vigorously for 1 h and left to settle for 12 h in a thermostatic bath (Lauda E200 water thermostat) at 298.15 K, to ensure complete separation between the two phases. Then, samples from both phases were taken with a syringe and their densities and refractive indices were measured at 298.15 K. In order to not disturb the equilibrium and avoid contamination from one phase with the other special capped needles were used. Lastly, the phase compositions in equilibrium were determined using the fitting of the densities and the refractive indices measured along the binodal curve with the composition, by means of the following equations:

$$\rho = Aw_1 + Bw_1^2 + Cw_1^3 + Dw_1^4 + Ew_2 + Fw_2^2 + Gw_2^3 + Hw_2^4 + Hw_2^4 + Iw_3 + Jw_3^2 + Kw_3^3 + Lw_3^4 \quad (2)$$

$$n_D = Mw_1 + Nw_1^2 + Ow_1^3 + Pw_1^4 + Qw_2 + Rw_2^2 + Sw_2^3 + Tw_2^4 + Uw_3 + Vw_3^2 + Ww_3^3 + Xw_3^4 \quad (3)$$

$$w_3 = 1 - (w_1 + w_2) \quad (4)$$

where w_1 , w_2 , w_3 correspond to the mass fraction of heptane, ethanol and ionic liquid, respectively, and A – X are adjustable

parameters. To evaluate the difference between the values given by the fitting equation and the experimental values, the root-mean-square-deviation (RMSD) was calculated through Eq. (5):

$$\text{RMSD} = \left(\sum_i^n \frac{(y_{\text{exp}} - y_{\text{calc}})^2}{n} \right)^{\frac{1}{2}} \quad (5)$$

where y_{exp} and y_{calc} are the experimental and calculated property, respectively, and n is the number of experimental points. The fitting parameters of Eqs. (2) and (3) together with the RMSD are shown in Tables S9 and S10 of the Supporting information (SI), for the system where the pure IL and the HIL was used, respectively.

3. Results and discussion

The density, viscosity and conductivity of the three ternary mixtures based on the $[\text{C}_2\text{MIM}][\text{SCN}]$ IL were measured and are presented in the SI. Note that $[\text{NH}_4][\text{SCN}]$ has a limited solubility in the $[\text{C}_2\text{MIM}][\text{SCN}]$ IL, and thus the measurement of the thermo-physical properties were done only up to $x[\text{NH}_4][\text{SCN}] = 0.1395$. The solubility of $[\text{NH}_4][\text{SCN}]$ in $[\text{C}_2\text{MIM}][\text{SCN}]$ was determined by the addition of very small amounts of the IS to the IL, followed by vigorous stirring until no further solid was present in the solution. This procedure was repeated until precipitation was observed even after 24 h of equilibrium. The measured density and viscosity data for the binary mixture $[\text{C}_2\text{MIM}][\text{SCN}] + [\text{NH}_4][\text{SCN}]$ as a function of inorganic salt molar fraction at different temperatures (298.15–353.15) K and at 0.1 MPa are given in Table S1 of the SI. The ionic conductivity of this mixture as a function of the $[\text{NH}_4][\text{SCN}]$ concentration in the temperature range from (298.15 to 323.15) K and at 0.1 MPa is listed in Table S2 of the SI.

Densities and viscosities of the two ionic liquid mixtures, $[\text{C}_2\text{MIM}][\text{SCN}] + [\text{C}_2\text{MIM}][\text{DCA}]$ and $[\text{C}_2\text{MIM}][\text{SCN}] + [\text{C}_2\text{MIM}][\text{C}_2\text{SO}_4]$ as a function of composition and in the temperature interval between (298.15 and 353.15) K and 0.1 MPa are given in Tables S3 and S4, respectively, of the SI. The ionic conductivities for both systems were measured at fixed temperatures in the interval between (298.15 and 323.15) K and are listed in Tables S5 and S6 for the $[\text{C}_2\text{MIM}][\text{SCN}] + [\text{C}_2\text{MIM}][\text{DCA}]$ and $[\text{C}_2\text{MIM}][\text{SCN}] + [\text{C}_2\text{MIM}][\text{C}_2\text{SO}_4]$, respectively, of the SI.

3.1. Ionicity

The classic Walden rule was used in order to determine the ionicity of the systems studied in this work. A plot of $\log(\Lambda_m)$ versus $\log(\eta^{-1})$ was made for the binary systems at different concentrations. For each concentration, different temperatures were plotted (from 298.15 K to 323.15 K). The molar conductivity, Λ_m , was calculated using the following equation:

$$\Lambda_m = \frac{\sigma}{\rho \times M} \quad (6)$$

with the conductivity, σ , in S cm^{-1} , the density, ρ , in g cm^{-3} and M the molar mass in g mol^{-1} . Using Eqs. (1) and (6) it was possible to provide the Walden plot for the studied systems. Figs. S1–S3 of the SI depicts the Walden plot for the studied systems $[\text{C}_2\text{MIM}][\text{SCN}] + [\text{NH}_4][\text{SCN}]$, $[\text{C}_2\text{MIM}][\text{SCN}] + [\text{C}_2\text{MIM}][\text{DCA}]$, and $[\text{C}_2\text{MIM}][\text{SCN}] + [\text{C}_2\text{MIM}][\text{C}_2\text{SO}_4]$, respectively. The ionicity of these mixtures was determined by evaluating the Walden deviation, ΔW , defined as the distance between the ideal Walden line and the experimental points, at a fixed fluidity. Fig. 1 shows a comparison between the ionicity of the three systems as a function of the molar composition of the component that was added to $[\text{C}_2\text{MIM}][\text{SCN}]$, at a constant fluidity of $\log(\eta^{-1}) = 0.5$. It can be observed that solubilising the inorganic salt, $[\text{NH}_4][\text{SCN}]$ in the pure $[\text{C}_2\text{MIM}]$

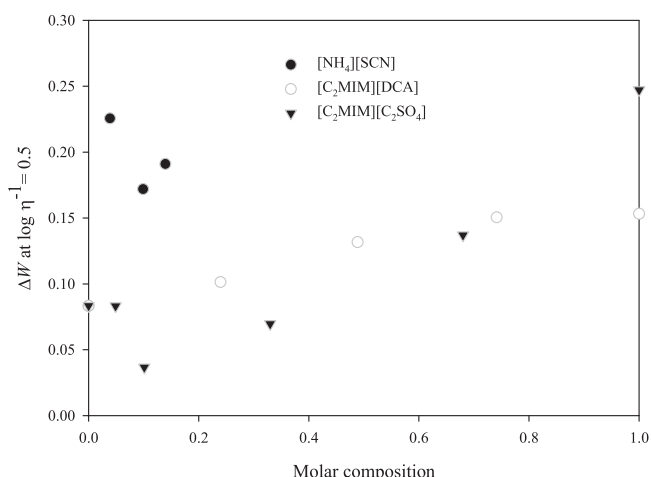


Fig. 1. Deviations from the “ideal” Walden line as a function of the molar fraction of component added to [C₂MIM][SCN].

[SCN] did not yield a system with increased ionicity, consequently not forming a HILL. The fact that the addition of the salt increased the viscosity of the pure [C₂MIM][SCN] and consequently, decreased the ionic conductivity might be behind the explanation for the mixtures presenting a lower ionicity than the pure IL. Note that both the IS and the IL have thiocyanate as the common anion, unlike other studies where [NH₄][SCN] was solubilised in ILs without a common anion, such as [C₂MIM][C₂SO₃] and [C₂MIM][C₂SO₄] [10], where the increase in ionicity was justified by the hydrogen bonds established between the [NH₄]⁺ cation and the anion of the IL, [C₂SO₃][−] or [C₂SO₄][−] leaving the [SCN][−] anion “free”. This suggests that the existence of a common anion may lead to lack of “competitiveness” between the ions involved in the system. However, the ionic liquid mixtures present ionicities that fall between those of the pure ILs, where pure [C₂MIM][SCN] shows a greater ionicity than that of [C₂MIM][DCA] and [C₂MIM][C₂SO₄]. Therefore, it could be interpreted that adding the thiocyanate based IL to the dicyanamide and ethylsulfate based ILs improves the ionicity of the latter ones. On the other hand, differences between the two systems can be observed: while for all the studied compositions of the [C₂MIM][DCA] containing system the corresponding ionicities fall between those of the pure fluids, displaying an ideal behaviour, for the [C₂MIM][C₂SO₄] containing system a more complex behaviour is observed. The addition of small amounts of [C₂MIM][C₂SO₄] to [C₂MIM][SCN] promotes an increase in the ionicity (decrease in the deviation to the ideal Walden line) regarding both pure fluids, reaching a minimum deviation when the mixture is composed of approximately 10% in molar fraction of [C₂MIM][C₂SO₄]. From this point onwards the ionicity decreases as the ethyl sulfate-based IL is added to the system, until the highest deviation of this system is reached, which corresponds to the pure [C₂MIM][C₂SO₄]. Therefore, it can be interpreted that, for this particular system, a mixture containing mole fraction of 0.1 of [C₂MIM][C₂SO₄] presents maximum ionicity, that is, the maximum mobility, or availability, of the ions.

To sum up, the solvent chosen to proceed to the liquid–liquid equilibria study was the mixture [C₂MIM][SCN]_{0.9}[C₂SO₄]_{0.1} as this mixture has proven to have a non-ideal behaviour that leads to an unexpected increase in the ionicity, falling into the HILL classification.

3.2. Liquid–liquid equilibria results and discussion

The experimental results for the liquid–liquid extractions of ethanol from the heptanes + ethanol azeotropic mixtures with both

the pure [C₂MIM][SCN] and the [C₂MIM][SCN]_{0.9}[C₂SO₄]_{0.1} HILL are presented in Table 2. Figs. 2 and 3 show the ternary diagrams for the systems heptane + ethanol + [C₂MIM][SCN] and heptane + ethanol + [C₂MIM][SCN]_{0.9}[C₂SO₄]_{0.1}, respectively, as well as the binodal curve and tie-lines. It can be observed that both systems present a fairly large immiscibility region and that the alkane and the ionic liquids are practically immiscible. The tie-lines obtained present positive slopes which imply that the distribution coefficients are higher than the unity and that the extraction of ethanol is favoured. The experimental data of the binodal curve for the systems heptane + ethanol + [C₂MIM][SCN] and heptane + ethanol + [C₂MIM][SCN]_{0.9}[C₂SO₄]_{0.1} are shown in Tables S7 and S8, respectively, of the SI.

Two parameters widely used to characterise the suitability of a solvent in a liquid–liquid extraction process are the distribution coefficient, β , and the selectivity, S , which are defined by Eqs. (7) and (8), respectively [13].

$$\beta = \frac{w_2^{\text{IL-rich phase}}}{w_2^{\text{HC-rich phase}}} \quad (7)$$

$$S = \frac{w_2^{\text{IL-rich phase}} w_1^{\text{HC-rich phase}}}{w_2^{\text{HC-rich phase}} w_1^{\text{IL-rich phase}}} \quad (8)$$

where w is the mass fraction, superscripts HC-rich phase and IL-rich phase indicate the hydrocarbon (top phase) and IL (bottom phase) rich phase, respectively, and subscripts 1 and 2 correspond to the heptane and ethanol, respectively.

The distribution coefficient describes the solute-carrying capability of the solvent (pure and mixture of ILs) and allows the determination of the amount of solvent required for the extraction process. The selectivity indicates the separation power of the solvent, indicating the ease of extraction of a certain solute (ethanol) from an inert (heptane). The ideal extraction solvent should present high values of both β and S , since high distribution coefficients imply a lower solvent flow rate, a smaller diameter column and consequently, a lower operating cost [13]. A high selectivity would decrease the number of equilibrium stages needed in an extraction unit and fewer amounts of inert residual would remain in the extract.

Table 2

Composition of the experimental phases in equilibrium for the ternary system heptane (1) + ethanol (2) + [C₂MIM][SCN] and [C₂MIM][SCN]_{0.9}[C₂SO₄]_{0.1} at 298.15 K and 0.1 MPa. The distribution coefficient, β , and the selectivity, S , are also shown.

{Azeotrope + [C ₂ MIM][SCN]}					
HC-rich phase		IL-rich phase		β	S
w_1	w_2	w_1	w_2		
0.9894	0.0094	0.0068	0.0785	8.33	1212.37
0.9876	0.0113	0.0104	0.1345	11.92	1135.65
0.9858	0.0134	0.0170	0.2632	19.60	1135.40
0.9594	0.0399	0.0272	0.3773	9.46	332.96
0.9292	0.0701	0.0430	0.4675	6.67	144.18
0.8741	0.1241	0.0765	0.5250	4.23	48.32
0.8107	0.1846	0.1295	0.5748	3.11	19.49
{Azeotrope + ([C ₂ MIM][SCN] + [C ₂ MIM][C ₂ SO ₄])}					
w_1	w_2	w_1	w_2	β	S
0.9902	0.0095	0.0021	0.1037	10.90	5149.33
0.9894	0.0098	0.0040	0.1989	20.20	4969.64
0.9788	0.0208	0.0149	0.2999	14.44	945.45
0.9599	0.0401	0.0241	0.3682	9.18	365.19
0.9366	0.0634	0.0450	0.4450	7.02	145.99
0.8913	0.1087	0.0748	0.5139	4.73	56.37
0.8325	0.1670	0.1200	0.5675	3.40	23.58

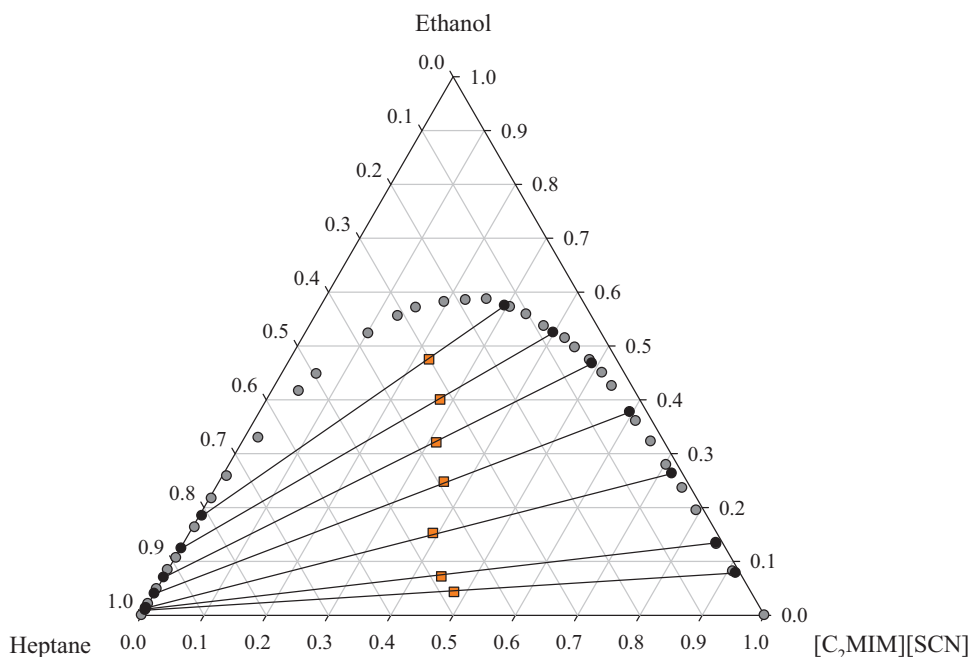


Fig. 2. Ternary diagram for the system heptane + ethanol + $[C_2MIM][SCN]$ at 298.15 K and 0.1 MPa. The grey dots represent the composition of the experimental binodal curve, the black dots the composition of the two phases in equilibrium and the orange squares the composition of the initial mixture. The lines connecting the composition of the two phases in equilibrium are the system tie-lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Figs. 4 and 5 show the selectivity and distribution coefficient, respectively, as a function of the ethanol mass fraction in the hydrocarbon-rich phase for both systems studied in this work. It can be observed that the $[C_2MIM][SCN]_{0.9}[C_2SO_4]_{0.1}$ HIL presents a selectivity about five times as high as the pure $[C_2MIM][SCN]$ IL at an ethanol mass fraction between 0.7% and 1.3% in the hydrocarbon-rich phase approximately. These values show that the $[C_2MIM][SCN]_{0.9}[C_2SO_4]_{0.1}$ HIL is more capable of extracting the ethanol from the heptane + ethanol azeotropic mixture than the

pure $[C_2MIM][SCN]$ IL. Slightly higher values of the distribution coefficients shown by the HIL can also be observed, indicating that it presents a greater solute-carrying capability than the pure $[C_2MIM][SCN]$. Moreover, the fact that the HIL is more selective than $[C_2MIM][SCN]$ at the referred concentration of solute, confirms that it has a greater extraction power, meaning that fewer equilibrium stages would be needed in the process and fewer amounts of inert residual would be present in the extract. Therefore, the maximum extraction is achieved when there is

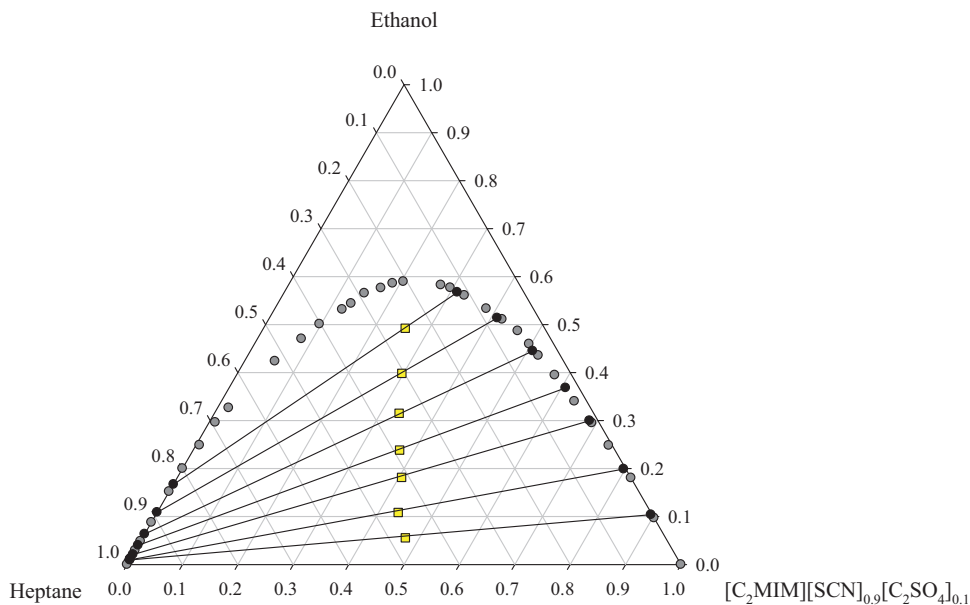


Fig. 3. Triangular diagram for the system heptane + ethanol + $[C_2MIM][SCN]_{0.9}[C_2SO_4]_{0.1}$ at 298.15 K and 0.1 MPa. The grey dots represent the composition of the experimental binodal curve, the black dots the composition of the two phases in equilibrium and the yellow squares the composition of the initial mixture. The lines connecting the composition of the two phases in equilibrium are the system tie-lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

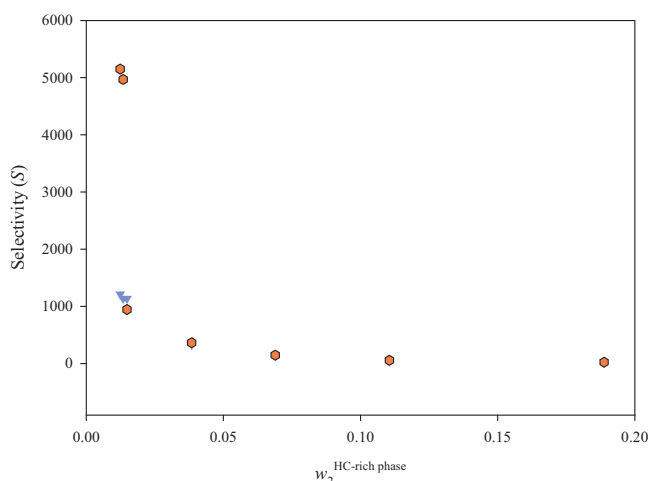


Fig. 4. Selectivity (S) of $[C_2MIM][SCN]$ (∇) and $[C_2MIM][SCN]_{0.9}[C_2SO_4]_{0.1}$ ($\bullet$) for the liquid–liquid extraction of ethanol from the heptane + ethanol system versus ethanol mass fraction in the hydrocarbon-rich phase ($w_2^{HC-rich phase}$) at 298.15 K and 0.1 MPa.

about 1 wt% of the solute in the hydrocarbon-rich phase, and a decrease is observed as more ethanol is present in the hydrocarbon-rich phase. This decrease is more abrupt at low concentrations of the alcohol in the upper phase, suggesting that the extraction of the solute towards the ionic liquid-rich phase is more favourable at low concentrations of ethanol.

Figs. 6 and 7 present a comparison in terms of selectivity and distribution coefficient, respectively, with the results obtained in this work and other ILs studied in the literature in the breaking of this azeotrope [1,14–16]. It is possible to observe that the $[C_2MIM][SCN]_{0.9}[C_2SO_4]_{0.1}$ is more selective than most of other ILs studied, with the exception of pure $[C_1MIM][C_1SO_4]$. It is possible to confirm that the presence of a shorter alkyl chain in the imidazolium cation leads to a higher selectivity [1]. The $[C_2MIM][SCN]_{0.9}[C_2SO_4]_{0.1}$ HIL studied in this work is among one of the ILs studied with the greatest distribution coefficient, as shown in Fig. 7. It can be interpreted that the presence of the specific proportion of $[C_2MIM][C_2SO_4]$ in the HIL is likely to play an

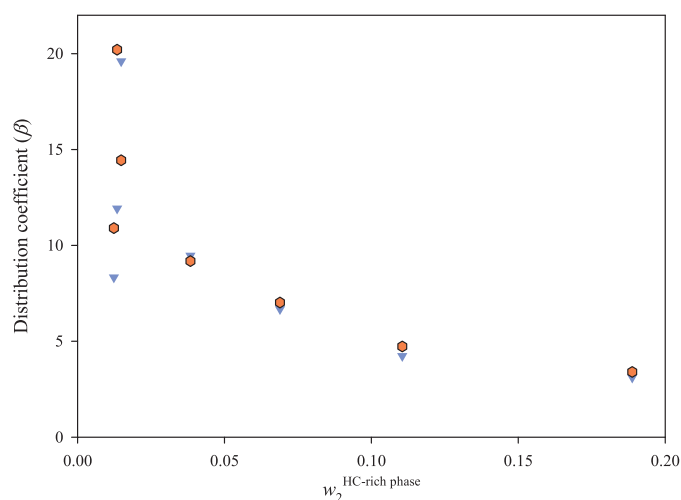


Fig. 5. Distribution coefficients (β) of $[C_2MIM][SCN]$ (∇) and $[C_2MIM][SCN]_{0.9}[C_2SO_4]_{0.1}$ ($\bullet$) for the liquid–liquid extraction of ethanol from the heptane + ethanol system versus ethanol mass fraction in the hydrocarbon-rich phase ($w_2^{HC-rich phase}$) at 298.15 K and 0.1 MPa.

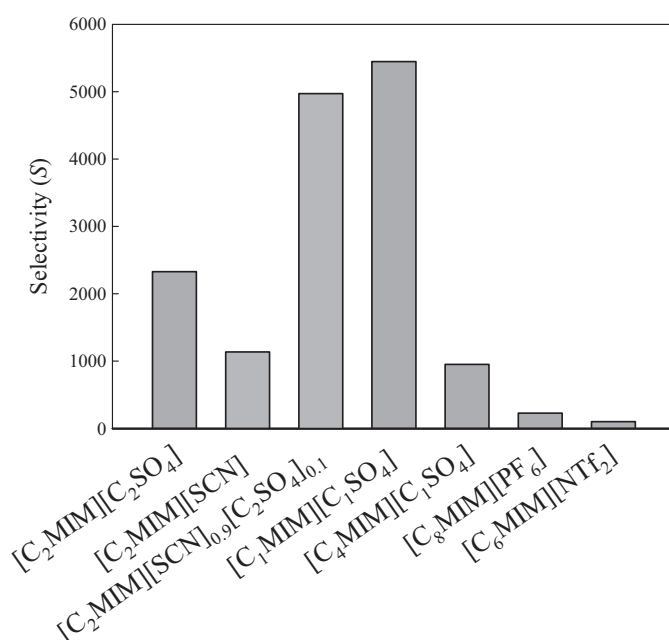


Fig. 6. Comparison of the selectivities (S) obtained in this work with those published in the open literature for other IL studied as solvents [12,14,15,17] for the separation of the azeotrope heptane + ethanol at an ethanol mass fraction of approximately 1 wt% in heptane-rich phase at 298.15 K and 0.1 MPa.

important role in terms of solute-carrying capacity of the solvent, where the sulfate anion can form stronger hydrogen bonding interactions with the ethanol than the other ILs. However, the fact that the solvent is composed mainly of $[C_2MIM][SCN]$ may also be behind the explanation in a somewhat subtle way. It was already established that the HIL has a greater ionicity than pure $[C_2MIM][SCN]$ approaching the behaviour of the ideal electrolyte. This means that the ions present in the system are less coordinated and

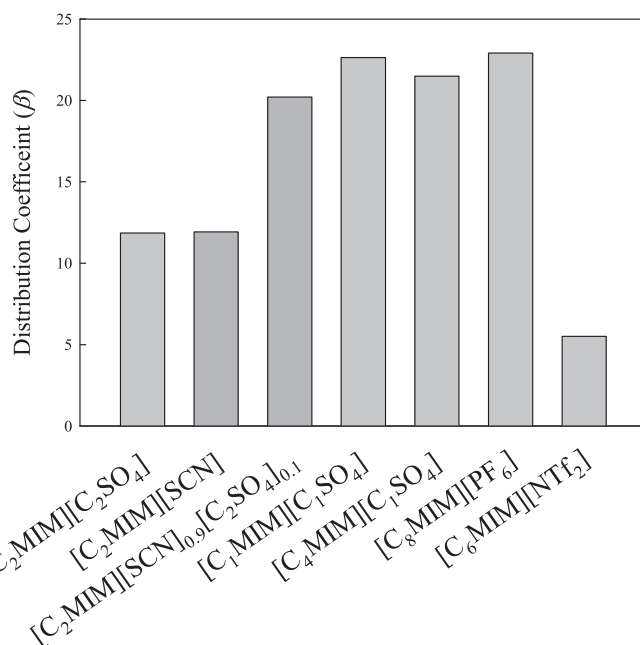


Fig. 7. Comparison of the distribution coefficients (β) obtained in this work with those published in the open literature for other IL studied as solvents [12,14,15,17] for the separation of the azeotrope heptane + ethanol at an ethanol mass fraction of approximately 1 wt% in heptane-rich phase at 298.15 K and 0.1 MPa.

more “free” to establish the possible interactions with the components involved. The thiocyanate anion is most probably the main responsible for the conduction process (pure [C₂MIM][SCN] has a high ionic conductivity already) but it is less likely to form H-bonds with the ethanol than the sulfate anion (oxygen is more electronegative than nitrogen). Therefore, the sulfate anion is more “free” to establish strong hydrogen bonds with the solute, consequently, justifying the enormous increase in the distribution coefficient.

4. Conclusions

The main goal of this work was to study the possibility of a relationship between the extraction efficiency of breaking the azeotropic mixture heptane+ethanol and the ionicity of the solvent used for the process. For this particular system, it was confirmed that increasing the ionicity of the pure ionic liquid [C₂MIM][SCN] by dissolving [C₂MIM][C₂SO₄] in a specific molar composition led to a more efficient and selective extraction of the alcohol. The selectivity and solute distribution ratio were observed to be higher than when choosing the pure ionic liquid as a solvent. Taking into account that the concept of ionicity is related to the ions available to participate in conduction and the fluidity of the system it may be reasonable to assume that the breaking of the azeotrope is more effective, owing to more “free ions” being present to establish interactions with the solute.

This work also permitted to demonstrate the ability of an ionic liquid as a solvent for the efficient separation of azeotropes by liquid–liquid extraction. The design of any extraction process should be addressed on the basis of: (i) determination of LLE data for the ternary systems and (ii) evaluation the extraction capacity with crucial parameters such as the distribution coefficients and selectivities. It can be concluded that the thiocyanate-based HIL, [C₂MIM][SCN]_{0.9}[C₂SO₄]_{0.1}, may be a good alternative to conventional solvents since it presented both high selectivity and distribution coefficient. In addition, it also allowed a raffinate purity of over 99 wt%, as shown by the experimental LLE data.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.fluid.2015.01.011>.

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